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Natural language processing in language learning: personalized and adaptive English language teaching using artificial intelligence

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Abstract

In recent years, artificial intelligence technology has begun to be widely used in the field of language teaching in colleges and universities. As an important branch of artificial intelligence, the study introduces natural language processing technology into English teaching in colleges and universities, utilizes it to extract rich semantic features in English teaching resources, combines LSTM and attention mechanism, and designs an English teaching resource recommendation model based on student's interests. Based on the recommendation model, an adaptive platform for English teaching is constructed. A small-scale trial is conducted to discuss students' feedback after the trial and explore the application effect of the platform. The results of each index of the English teaching resources recommendation model are optimal, and the recall (88.65%), accuracy (90.29%) and NGDD (0.3725) are higher than those of other models, which proves the model's effectiveness in resource recommendation. The positive feedback from students on the adaptive effect, applicability, and recognition of the adaptive platform for English teaching is basically above 50%. The analysis reveals that the AI-based resource recommendation model and adaptive English teaching can aid English teaching in colleges and universities and enhance the quality of English teaching.

Keywords: LSTM; Attention mechanism; Resource recommendation; Natural language processing; English teaching.
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1 Introduction

In recent years, artificial intelligence AI has made breakthroughs in many fields, such as natural language understanding and image recognition. How to introduce AI technology into college English teaching and efficiently integrate the “teaching” and “learning” aspects of college English has not been fully studied.

Adaptive teaching is a kind of efficient teaching method to teach knowledge and skills to learners, which can make learners make progress and is an effective strategy to make up for the lack of learners' ability [1-2]. Natural language processing, as a kind of big data technology, can transform the language used in human communication into machine-understandable language after processing, which is a digital tool for realizing personalized English teaching [3-6]. The language proficiency model and algorithm framework supported by it is the intersection of linguistics and computer science knowledge, which can meet the rigid needs of English learning for context and effectively strengthen learners' independent learning ability in English “listening”, “speaking”, “reading”, and “writing” [7-9]. At present, it is possible to develop software based on speech recognition technology that can realize the practice functions of “listening” and “speaking” in English teaching, and the functions included in the software include general dialogue, voice search, intelligent assistance, language learning, etc., to promote learners' interest in “listening” and “speaking” and the quality of practice [10-12]. Branch techniques such as word segmentation, word vector modeling, word attribute labeling, sentence breaking, and sentence meaning segmentation under natural language processing technology play a huge role in the recognition, analysis, and comprehension of texts as a whole and effectively improve learners' ability to read materials and write in English [13-15].

Alrawashdeh, G. S. et al. showed that personalized and adaptive learning approaches were more effective than traditional curriculum approaches in improving learners' reading achievement and evaluated the impact in the context of a broader population and intervention [16]. Vaughn, M., guided by social constructivism and teachers' metacognitive theories and based on empirical investigations, examined adaptive reading instruction with respect to students' teaching needs and specific implementation paths [17]. Kacetl, J. et al. illustrated the important utility of mobile apps for English language learning, such as significantly enhancing learners' cognitive ability, motivation, autonomy and self-confidence and facilitating their personalized learning, but the reasonable instructional design is also the key to helping learners achieve their learning goals [18]. Ruan, S. et al. investigated the role of educational chat Ruan, S. et al. studied the role of educational chatbots in English language teaching, and compared with the traditional media for learning factual knowledge, the proposed QuizBot conversational intelligence can attract students' interest in learning and has better learning facilitation effects [19]. Pandarova, I. et al. proposed an intelligent English tense teaching system that dynamically adjusts the difficulty of learning content according to the learners' language ability and creates an adaptive language learning environment for learners with different individual characteristics [20]. Hao, Y. et al. designed an English learning mobile application based on the framework of cognitive apprenticeship, which provides learners with evidence-based learning materials for learning English as a foreign language. Hao, Y. et al. designed an English learning mobile application based on the framework of cognitive apprenticeship, which provides learners with evidence-based learning materials for learning English as a foreign language and effectively improves learners' adaptability in learning English [21]. Nagata, R. et al. designed a spelling autocorrection method to address the problem of learners' English spelling errors, which involves creating a spelling correction model to the original learner corpus in order to minimize the impact of grammatical and spelling errors on the performance of the English Natural Language Processing (NLP) system [22].

In this paper, we make a theoretical analysis of the application of natural language processing technology in language learning, analyze the reader's interest by using the long and short-term

memory network model, and introduce the attention mechanism to make effective interaction between learning resources and students, so as to put forward a learning resource recommendation model based on students' interest for English teaching. Comparison experiments are conducted on three dimensions of model recommendation performance, teaching quality, and operational efficiency to analyze the model's resource recommendation effect and its auxiliary effect on English teaching. Combined with the constructed resource recommendation model, an adaptive platform for personalized recommendation of English learning resources has been designed. Students of a middle school are selected as the experimental subjects for the trial of teaching on the adaptive platform, and after the experiment, data are obtained through questionnaires to analyze the adaptive effect of the platform, the use effect and the degree of recognition, and to assess the application effect of the adaptive platform for English teaching.

2 Application of natural language processing techniques to language learning

Natural Language Processing (NLP) technology is a key branch in the field of Artificial Intelligence, aiming to enable computers to understand, process, and generate human language. The use of natural language processing technology opens up new opportunities for transforming education models in colleges and universities. The application of natural language processing technology in language learning is shown in Figure 1, including aspects of promoting personalized education, multimodal intelligent teaching, and optimizing education management.

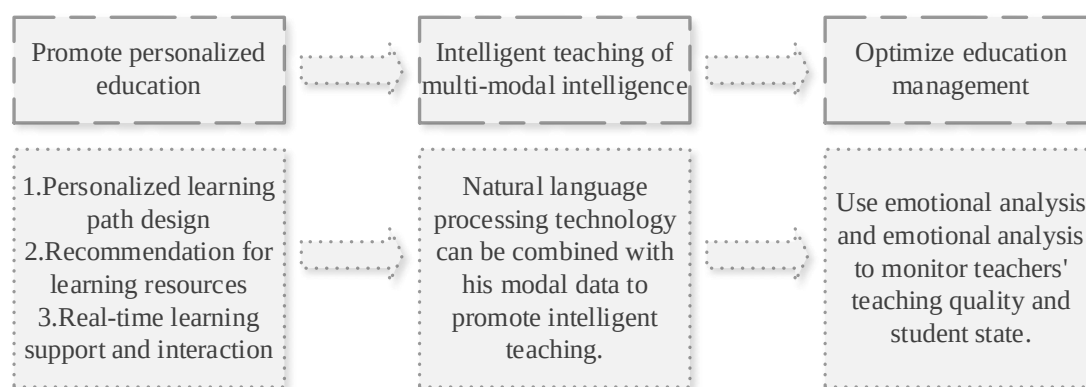


Figure 1. The application of natural language processing technology in linguistics

2.1 Promoting personalized education

Artificial intelligence will play a significant role in personalized education in the future, and natural language processing technology is a powerful tool for achieving personalized education.

1) Personalized Learning Path Design

Based on the comprehensive analysis of students' subject interest, learning speed and ability, natural language processing technology can be used to assist educators in designing personalized language learning paths so that each student can learn according to their learning rhythm and way and maximize the effect of tailoring education to the student's needs.

2) Learning Resources Recommendation

Through the text analysis and sentiment analysis of natural language processing technology, we analyze the text data generated by the students in their learning and gain a deeper

understanding of their understanding of different knowledge points and their interest in learning and other information. Based on this information, personalized language learning resources are recommended for them, including textbooks, courses, and learning applications. The efficiency of students' knowledge acquisition is improved and the language-learning process becomes more interesting and fulfilling.

3) Real-time Learning Support and Interaction

The application of natural language processing technology establishes a real-time learning support system to provide students with personalized and timely language learning assistance. The system analyzes students' language text data to understand their confusion about specific knowledge points. It offers students a platform for real-time feedback and interaction that fosters closer communication and cooperation between teachers and students. This real-time learning support and interaction mechanism will significantly improve students' academic self-confidence and learning effectiveness, and provide more comprehensive support for personalized education.

2.2 Multimodal Intelligent Teaching

Natural language processing technologies can be combined with other modal data (e.g., vision, voice) to drive intelligent teaching and learning. The future of education will rely on multimodal technologies such as face, voice, motion, emotion, and eye recognition to build a series of intelligent teaching support systems. These systems will not only monitor students' attendance, classroom performance, and concentration, but also assess the quality of teachers' teaching and generate timely classroom assessment reports to provide feedback. Teachers can adjust the content and mode of instruction based on an intelligent assessment to ensure a high-quality classroom experience, forming a closed loop of comprehensive monitoring that combines behavior, recognition, and analysis.

Under this trend, the presentation format of learning resources will be more diverse, no longer limited to traditional text-based learning products. With the help of natural language processing technology, learning resources will include text, voice, image, video, and other multimodal learning materials. Students can more easily choose the resources that suit them, thus improving the efficiency and depth of learning.

2.3 Optimizing education management

Sentimental analysis and sentiment analysis are important applications of natural language processing technology. Universities can use applications like sentiment analysis and emotion analysis to monitor school public opinion in real-time, assess the quality of teachers' teaching and students' rationale status, and more. This aids in improving teaching quality, intervening in students' language learning issues in a timely manner, and providing data support for school management to optimize decision-making and resource allocation.

Starting from personalized and adaptive English teaching, this paper constructs an English learning resource recommendation model based on natural language processing technology to illustrate the practical application of AI technology in language teaching.

3 Model for recommending learning resources for English language teaching and learning

Deep learning has become the hottest research direction in recent years due to its strong expressive ability and flexible model structure, and this paper uses it in English teaching resources recommendation and proposes an English teaching resources recommendation model based on students' interest mining in response to the problems of data sparsity and weak generalization ability of traditional recommendation algorithms.

3.1 LSTM networks

LSTM, also known as Long Short-Term Memory Network, is an improved model of the basic recurrent neural network RNN, which introduces three “gate” units, the input gate, the forget gate and the output gate, to decide whether to discard or add information.

In LSTM network structure, information runs through from input gate i_t to output gate o_t , which is like a conveyor belt passing information from the previous unit to the next unit, thus realizing the function of forgetting or remembering, and therefore better alleviating the gradient vanishing problem of RNN. The computational process is shown in Eq. (1)-Eq. (6):

$$f_t = \text{sigmoid}(W_{xf}x_t + W_{hf}h_{t-1} + b_f) \quad (1)$$

$$i_t = \text{sigmoid}(W_{xi}x_t + W_{hi}h_{t-1} + b_i) \quad (2)$$

$$\tilde{C}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \quad (3)$$

$$C_t = f_t \circ C_{t-1} + i_t \circ \tilde{C}_t \quad (4)$$

$$o_t = \text{sigmoid}(W_{xo}x_t + W_{ho}h_{t-1} + b_o) \quad (5)$$

$$h_t = O_t \circ \tanh(C_t) \quad (6)$$

In Eq. (1)-Eq. (6), f_t , i_t , and o_t represent the forgetting gate, input gate, and output gate, respectively, sigmoid and tanh represent the activation function, $\circ$ represents the element-wise multiplication, x_t is the input at the current moment, h_{t-1} is the hidden state at the previous moment, $\tilde{C}_t$ is the new state obtained after the tanh, C_{t-1} is the unit state at the previous time, C_t is the updated state, and h_t is the final output. W_{xf} , W_{hf} and b_f represent the weight matrices and bias terms of x_t , h_{t-1} in the forgetting gate, W_{xi} , W_{hi} and b_i represent the weight matrices and bias terms of x_t , h_{t-1} in the input gate, W_c and b_c represent the weight matrices and bias terms of $\tilde{C}_t$, W_{xo} , W_{ho} and b_o represent the weight matrices and bias terms of x_t , h_{t-1} in the output gate, respectively.

3.2 Attention mechanism

Attention mechanism, i.e., attention mechanism, is inspired by the imitation of human visual attention. Firstly, the correlation function calculates the correlation between the input sequence Key (key) and

the target task Query (query), and after normalization, the attention weight coefficients of each sequence are obtained, and the calculated weight coefficients are weighted and summed up with the corresponding Value values of Key to get the final attention weights.

3.3 Resource Recommendation Model

The DRPM model based on student interest mining proposed in this study is mainly constructed by the input layer, feature extraction layer, preference extraction layer, and MLP layer. First, student borrowing history, candidate resources, and reader demographic features are inputted into the model. The text features and reader features of learning resources are extracted by the feature extraction layer. Then, the extracted vectors of student borrowing resources are inputted into the preference extraction layer to mine students' reading interests. Finally, the weighted history features, candidate resource features, and student features are combined and inputted into the multilayer perceptual machine to obtain the recommendation results.

3.3.1 Input Layer

The model input layer is information that can express the characteristics of students and English learning resources, including attributes such as ID number, unit, gender, and category, as well as content features such as book titles and summaries. In addition, in order to alleviate the user cold start problem, the textbook information of each semester is associated with the borrowing records of relevant people by unit and grade group so that the model can provide relevant learning resource recommendations based on the professional background of new readers when they lack historical borrowing behavior.

3.3.2 Feature extraction layer

The BERT model is used to extract the semantic features of students' borrowing history and candidate resources, resulting in a vector of student borrowing behaviors and a vector of candidate resources.

3.3.3 Preference Extraction Layer

The preference extraction layer learns student preferences by analyzing student behavior to mine reader interests. First, the student borrowing resource vectors extracted in the feature extraction layer are input into the LSTM network for training, and then the trained feature vectors interact with the candidate resources, and then the different attention weights of the English learning resources are computed to get the feature vectors of the learning resources that conform to the student's interest preferences.

For the features extracted from the LSTM model, different weights are assigned to students' borrowing behaviors through the mechanism of increasing attention, and the more relevant learning resources are assigned higher coefficients, so as to differentiate the importance of the learning resources in the borrowing records, and to realize the dynamic extraction of borrowing behavior preferences. The calculation process is shown in equations (7) and (9):

$$\text{sim}(q_j, k) = k^T q_j \quad (7)$$

$$a_j = \text{soft max} \left(\text{sim}(q_j, k) \right) = \frac{\exp(\text{sim}(q_j, k))}{\sum_{j=1}^n \exp(\text{sim}(q_j, k))} \quad (8)$$

$$c = \sum_{j=1}^n a_j q_j \quad (9)$$

Where k^T represents the transpose of the representation vector k of the recommended resources and q represents the vector representation of the j th resource borrowed by the student. The result $\text{sim}(q_j, k)$ obtained after the inner product of the two is exponentially normalized using the soft max function to compute the attention weight score a_j with respect to the candidate resource; c represents the final student preference vector, and n represents the total number of student borrowings.

3.4 Experimentation and analysis of the model

To verify the feasibility of the personalized recommendation model of English teaching resources based on students' interests, the following experiments have been designed.

3.4.1 Experimental setup

The experiments use the English reading materials in a university library as English teaching resources to test the application performance of the model, and the evaluation indexes include the recall rate R, the accuracy rate P, and the normalized discounted cumulative gain NDGG. In order to avoid the experimental results being too homogeneous, the model of this paper, the Item CF model, the PMF model, the BERT-MLP model, and the BERT-LSTM-MLP model are tested in comparison with one another. Comparison test.

3.4.2 Performance validation

The model recommendation performance metrics test results are shown in Figure 2. Compared with the other four recommendation models, the index values of recall, accuracy and normalized discounted cumulative gain of this paper's model are larger, respectively 88.65%, 90.29% and 0.3725, indicating that this paper's recommendation model has a better effect in recommending English teaching resources.

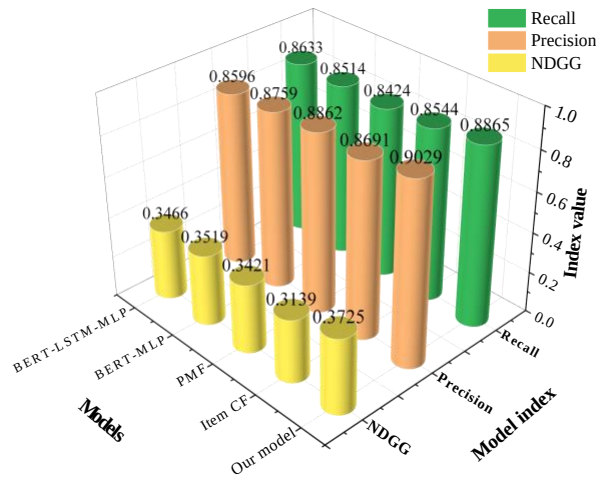


Figure 2. Model recommendation test results

3.4.3 Quality of teaching

Due to the large differences between different teaching resource recommendation models, the final quality of teaching is also different. The experiment selected 30 English majors with close grades as the test object, divided into 10 different groups, every 3 students as a group, respectively, selected this paper's model, Item CF model, PMF model of three different teaching resource recommendation model to assist teaching, and finally tested, The goodness of the teaching model is determined by the students' scores, in which the full score is 100 points, 100 points to 85 points is excellent, 84 points to 60 points is passing, and less than 60 points is failing.

Figure 3 illustrates the comparison results of the teaching quality of various teaching models. The three resource recommendation models got different test results, although all of them reached excellent, the students' scores of this paper's resource recommendation model are above 90 points, which is significantly higher than the other two resource recommendation models, proving the effectiveness and feasibility of this paper's resource recommendation model for recommending English teaching resources.

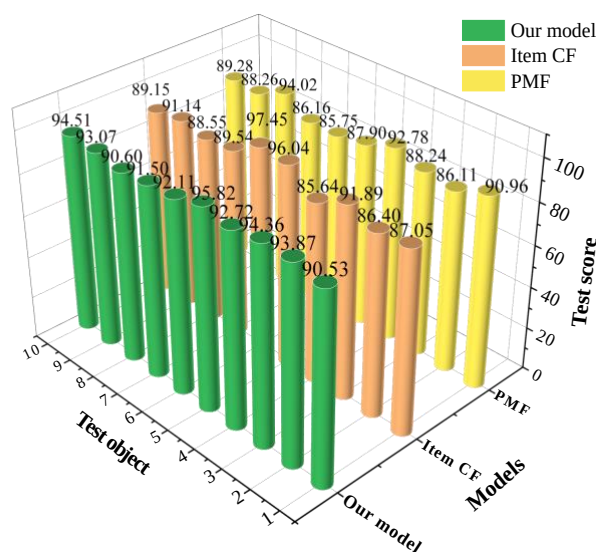


Figure 3. The comparison of the teaching quality of different teaching models

3.4.4 Operational efficiency

The results of the running efficiency comparison of different teaching models are shown in Figure 4. Under different numbers of student tests, the running efficiency of the English teaching resource recommendation model in this paper is 82.58%~94.86%, and its average running efficiency with the three teaching resource recommendation models of Item CF and PMF is 88.80%, 84.00% and 87.12%, respectively, and the English teaching resource recommendation model based on students' interests has a higher running efficiency.

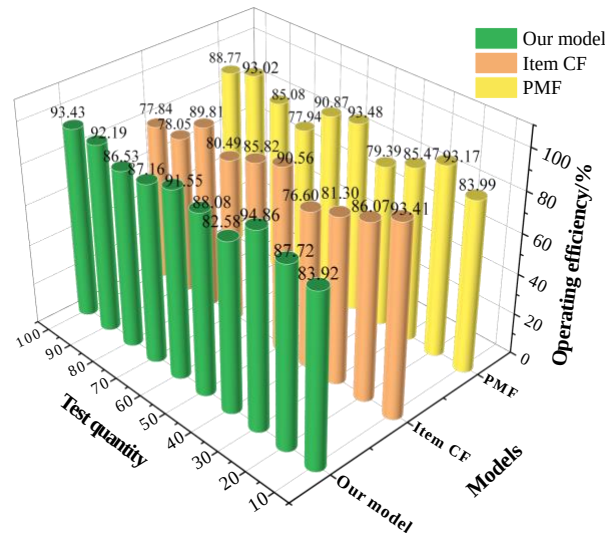


Figure 4. Comparison results of the operating efficiency of different teaching models

4 Adaptive platform and applications for English language teaching and learning

The adaptive learning platform in this study is primarily utilized by English language teaching students, with the objective of constructing an adaptive learning environment that combines a perfect learning resource platform with personalized recommendations. The main objective of this adaptive system is to suggest personalized learning resources to student users based on their data.

4.1 System module design

The adaptive learning system designed in this study mainly focuses on learning English as a subject through the recommendations of English learning resources. The module design of the adaptive platform for English teaching is shown in Figure 5. The system obtains information about students' learning behavior data, including personal information and learning interests, and pushes personalized English learning resources for students on the basis of analyzing their personal information and learning degree level so as to improve the efficiency of students' learning to a greater extent.

The system is subdivided into a student module and an administrator module. In the student module, there are three modules: a student registration and login module, a student interest assessment module, a results query module, a personalized learning resources recommendation module, and a personal information management module. The administrator module includes login, user information management, learning resource management, and assessment module management.

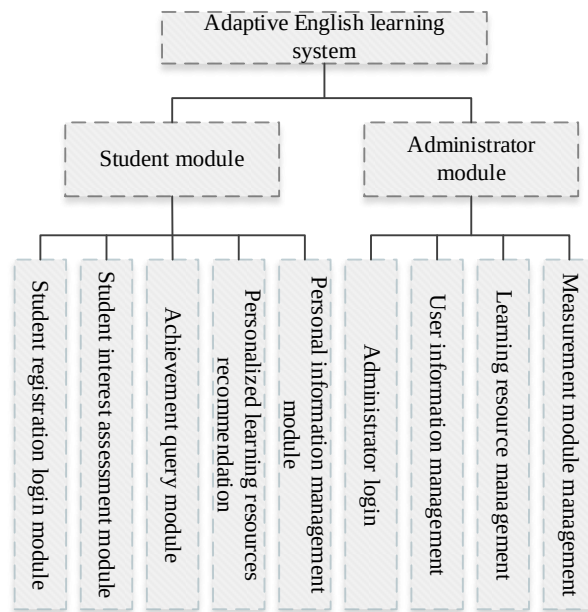


Figure 5. The design of the adaptive platform of English teaching

4.2 Investigation and analysis of application effects

To verify whether this adaptive learning platform for teaching English improves the quality of students' learning during the learning process, a feedback survey was conducted among the users. Feedback from users was collected during a small scale testing of the platform with students from a secondary school.

The study collected feedback from students' information by using questionnaires and analyzed the questionnaire statistics to understand the effectiveness of the platform's use. The questionnaire surveyed the intelligent learning platform from three dimensions, namely, users' personal information, learners' effect of using the platform, and learners' evaluation of the platform's recognition level, consisting of 11 questions in total. By conducting a survey and analysis of these three dimensions, it is possible to evaluate both the effectiveness of the platform function and the rationality of the platform design. After the system is online, questionnaires are distributed to system users to conduct surveys.

4.2.1 Users' personal information

The total number of specific subjects was 300, the valid questionnaires were 276, and the validity rate of the questionnaires was 92%. Of these, 54.34% were female learners and 45.66% were male learners.

4.2.2 Investigation of adaptive effects

Adaptive learning refers to the ability to arrange learning strategies and resources for students that are suitable for individual learners according to their learning characteristics and preferences. A survey was conducted to investigate whether the platform satisfies adaptive learning, and the results of the survey on the adaptive effect of the platform are shown in Table 1. The cumulative percentage of subjects who think that they are very satisfied and relatively satisfied is 64.9%, and the cumulative percentage of the first three items is 80.4%, which leads to the conclusion that the platform satisfies adaptive learning in English teaching.

Table 1. The results of the platform's adaptive effect

		Frequency	Percentage	Effective percentage	Cumulative percentage
In effect	Very satisfied	52	17.3%	18.8%	18.8%
	Quite satisfied	127	42.3%	46.0%	64.9%
	Generally	43	14.3%	15.6%	80.4%
	Not too satisfied	33	11.0%	12.0%	92.4%
	Not satisfied	21	7.0%	7.6%	100.0%
	Total	276	92.0%	100.0%	
Loss	System	24	8.0%		
Total		300	100.0%		

4.2.3 Survey of the effectiveness of the platform's use

The resource recommendation model is designed to recommend English learning resources based on student interests, so the student interest test examination must be comprehensive and accurate. The results of the survey on the comprehensiveness of the students' interest examination are shown in Table 2. The results statistically showed that 50.4% of the students thought that the test results were comprehensive and accurate, 23.2% thought that the test results were inaccurate, and the overall effect of the student interest test was perfect.

Table 2. The comprehensiveness of the interest of the students

		Frequency	Percentage	Effective percentage	Cumulative percentage
In effect	Very sufficient	62	20.7%	22.5%	22.5%
	Quite sufficient	77	25.7%	27.9%	50.4%
	Generally	73	24.3%	26.4%	76.8%
	Not too sufficient	52	17.3%	18.8%	95.7%
	Not sufficient	12	4.0%	4.3%	100.0%
	Total	276	92.0%	100.0%	
Loss	System	24	8.0%		
Total		300	100.0%		

The personalized recommendation function is the focus of adaptive learning in English teaching. A questionnaire survey on whether personalized recommendation meets the needs of students and the degree of compliance of recommendation needs is shown in Table 3. After data analysis of the personalized recommendation needs, which accounted for 47.5% of the students' personalized recommendation needs, the results show that the adaptive system is more in line with the students' personalized English learning.

Table 3. The degree of compliance of recommended requirements

		Frequency	Percentage	Effective percentage	Cumulative percentage
In effect	Very conforming	53	17.7%	19.2%	19.2%
	Quite conforming	78	26.0%	28.3%	47.5%
	Generally	66	22.0%	23.9%	71.4%
	Not too conforming	52	17.3%	18.8%	90.2%
	Not conforming	27	9.0%	9.8%	100.0%
	Total	276	92.0%	100.0%	
Loss	System	24	8.0%		
Total		300	100.0%		

4.2.4 Survey on platform recognition

Through the use of the Adaptive Platform for Teaching English as a Foreign Language, the degree of users' recognition of the platform was investigated. The overall functional satisfaction of the adaptive platform was investigated, and the satisfaction level of the overall functionality of the platform is shown in Table 4. Subjects were more satisfied with the overall functional statistics of the Adaptive Platform for English Language Teaching, and the cumulative percentage of those who were very satisfied and more satisfied was 52.5%, while those who were less satisfied and dissatisfied were around 20%.

Table 4. The overall function satisfaction of the platform

		Frequency	Percentage	Effective percentage	Cumulative percentage
In effect	Very satisfied	42	14.0%	15.2%	15.2%
	Quite satisfied	103	34.3%	37.3%	52.5%
	Generally	72	24.0%	26.1%	78.6%
	Not too satisfied	41	13.7%	14.9%	93.5%
	Not satisfied	18	6.0%	6.5%	100.0%
	Total	276	92.0%	100.0%	
Loss	System	24	8.0%		
Total		300	100.0%		

5 Conclusion

With the progress of science and technology and the development of natural language processing, it has become feasible to utilize computers for English teaching assistance. The study utilizes artificial intelligence technology to construct a learning resource recommendation model for English teaching and design an adaptive platform for English teaching based on the resource recommendation model and investigates the application effects of the learning resource recommendation model and the adaptive English teaching platform through the model comparison experiments and teaching experiments.

In terms of recommending English teaching resources based on student's interests, the recall rate (88.65%), accuracy rate (90.29%) and NGDD (0.3725) of the English reading materials recommended by this paper's model are higher than those of the comparison model, and the teaching quality scores of using this paper's model to assist English teaching are all greater than 90, which reaches the excellent level. This paper's model presents the best results in the comparative analysis of the three dimensions of recommendation performance, teaching quality, and operational efficiency.

After using the adaptive platform for English teaching, 64.9% and 52.5% of the students were satisfied with its adaptive learning effect and approved of the overall function, respectively, and 50.4% and 47.5% thought that the interest test was comprehensive and accurate and the personalized recommendation met the needs. On the whole, learners are basically satisfied with the application effect of the adaptive platform for English teaching, and the platform still needs to be strengthened in processing test results and personalized recommendations.

Artificial intelligence technology has brought many innovations and changes to English teaching in colleges and universities, which can provide personalized learning experience, real-time feedback and assessment, automated generation and assessment of topics and many other functions to help

students master knowledge and skills faster. However, AI technology also has some limitations. Therefore, when applying AI technology to English teaching in colleges and universities, it is necessary to consider various factors and find the best application methods and strategies to achieve the best teaching effect and experience.

References

- [1] Frydland, N. (2017). Implementing the mutually adaptive learning paradigm: Applications for literacy-level through academic ESL. *College ESL Quarterly*, 1, 6.
- [2] Rizvi, S., Rienties, B., Rogaten, J., & Kizilcec, R. (2022). Culturally adaptive learning design. *Open World Learning*, 103(1), 39-43.
- [3] El-Sabagh, H. A. (2021). Adaptive e-learning environment based on learning styles and its impact on development students' engagement. *International Journal of Educational Technology in Higher Education*, 18(1), 53.
- [4] Shazeer, N., & Stern, M. (2018, July). Adafactor: Adaptive learning rates with sublinear memory cost. In *International Conference on Machine Learning* (pp. 4596-4604). PMLR.
- [5] Salguero-Serrat, J., Sánchez, H., Naranjo, M. J., Rico, M., & Agudo, J. E. (2019, June). STVALL: HbbTV Based Adaptive System for English Learning through Interactive TV. In *EdMedia+ Innovate Learning* (pp. 1520-1525). Association for the Advancement of Computing in Education (AACE).
- [6] Pokrivcakova, S. (2019). Preparing teachers for the application of AI-powered technologies in foreign language education. *Journal of Language and Cultural Education*, 7(3), 135-153.
- [7] Sottolare, R. A., & Goodwin, G. A. (2017). Adaptive instructional methods to accelerate learning and enhance learning capacity. In *International Defense & Homeland Security Simulation Workshop of the I3M Conference*.
- [8] Ciloglul, B., & Inceoglu, M. M. (2018). A learner ontology based on learning style models for adaptive e-learning. In *Computational Science and Its Applications–ICCSA 2018: 18th International Conference, Melbourne, VIC, Australia, July 2-5, 2018, Proceedings, Part II 18* (pp. 199-212). Springer International Publishing.
- [9] Du, Y., Wang, J., Feng, W., Pan, S., Qin, T., Xu, R., & Wang, C. (2021, October). Adarnn: Adaptive learning and forecasting of time series. In *Proceedings of the 30th ACM international conference on information & knowledge management* (pp. 402-411).
- [10] Adnan, M., Habib, A., Ashraf, J., & Mussadiq, S. (2019). Cloud-supported machine learning system for context-aware adaptive M-learning. *Turkish Journal of Electrical Engineering and Computer Sciences*, 27(4), 2798-2816.
- [11] Huang, J., Saleh, S., & Liu, Y. (2021). A review on artificial intelligence in education. *Academic Journal of Interdisciplinary Studies*, 10(3).
- [12] Parsons, S. A., Vaughn, M., Scales, R. Q., Gallagher, M. A., Parsons, A. W., Davis, S. G., ... & Allen, M. (2018). Teachers' instructional adaptations: A research synthesis. *Review of educational research*, 88(2), 205-242.
- [13] Liang, J., Hare, R., Chang, T., Xu, F., Tang, Y., Wang, F. Y., ... & Lei, M. (2022). Student modeling and analysis in adaptive instructional systems. *IEEE Access*, 10, 59359-59372.
- [14] Othmane, Z. I. N. E., DEROUICH, A., & TALBI, A. (2019). A comparative study of the Most influential learning styles used in adaptive educational environments. *International Journal of Advanced Computer Science and Applications*, 10(11).
- [15] Fontaine, G., Cossette, S., Maheu-Cadotte, M. A., Mailhot, T., Deschênes, M. F., & Mathieu-Dupuis, G. (2017). Effectiveness of adaptive e-learning environments on knowledge, competence, and behavior in health professionals and students: protocol for a systematic review and meta-analysis. *JMIR research protocols*, 6(7), e8085.

- [16] Alrawashdeh, G. S., Fyffe, S., Azevedo, R. F., & Castillo, N. M. (2023). Exploring the impact of personalized and adaptive learning technologies on reading literacy: A global meta-analysis. *Educational Research Review*, 100587.
- [17] Vaughn, M. (2019). Adaptive teaching during reading instruction: A multi-case study. *Reading Psychology*, 40(1), 1-33.
- [18] Kacetl, J., & Klímová, B. (2019). Use of smartphone applications in english language learning—A challenge for foreign language education. *Education Sciences*, 9(3), 179.
- [19] Ruan, S., Jiang, L., Xu, J., Tham, B. J. K., Qiu, Z., Zhu, Y., ... & Landay, J. A. (2019, May). Quizbot: A dialogue-based adaptive learning system for factual knowledge. In *Proceedings of the 2019 CHI conference on human factors in computing systems* (pp. 1-13).
- [20] Pandarova, I., Schmidt, T., Hartig, J., Boubekki, A., Jones, R. D., & Brefeld, U. (2019). Predicting the difficulty of exercise items for dynamic difficulty adaptation in adaptive language tutoring. *International Journal of Artificial Intelligence in Education*, 29, 342-367.
- [21] Hao, Y., Lee, K. S., Chen, S. T., & Sim, S. C. (2019). An evaluative study of a mobile application for middle school students struggling with English vocabulary learning. *Computers in Human Behavior*, 95, 208-216.
- [22] Nagata, R., Takamura, H., & Neubig, G. (2017). Adaptive spelling error correction models for learner English. *Procedia Computer Science*, 112, 474-483.